10.1 Surface Area

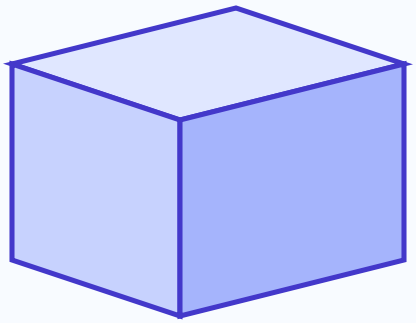
Surface area (SA) is the total area of every face of a 3D solid — the amount of material needed to wrap it completely. Prisms and cylinders use $SA = 2B + P\,h$ (or the equivalent $2(lw + lh + wh)$ for a box), pyramids and cones use a base plus *lateral* area built from the **slant height** ℓ (not the vertical height), and $SA = 4\pi r^2$ for a sphere. Many real problems hide a Pythagorean-theorem step: the slant height of a cone or pyramid is rarely given outright — you often build it from the height and a base measurement first. This worksheet also covers composite solids (two shapes fused, with the shared seam excluded) and reverse problems where SA is known and a dimension is missing.

Examples

Example 1: Rectangular prism — full steps

Find the surface area of a rectangular prism with length 8, width 5, and height 6.

Solution



A box has 3 pairs of congruent faces: top/bottom, front/back, left/right.

A box has three pairs of congruent rectangular faces, so $SA = 2(lw + lh + wh)$.

$lw = 8(5) = 40,\quad lh = 8(6) = 48,\quad wh = 5(6) = 30.$

$SA = 2(40 + 48 + 30) = 2(118) = 236.$

Answer: 236 square units.

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Example 2: Cube

Find the surface area of a cube with edge length 7.

Solution

A cube has 6 congruent square faces, so $SA = 6s^2$.

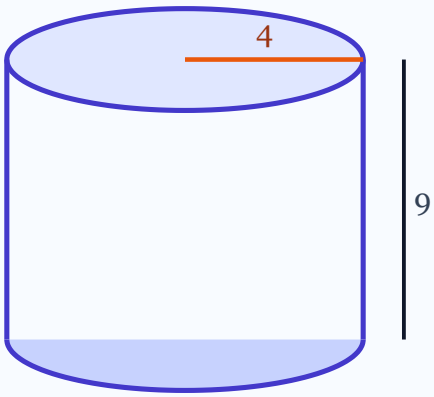
$SA = 6(7^2) = 6(49) = 294$.

Answer: 294 square units.

Example 3: Cylinder — full steps

Find the total surface area of a cylinder with radius 4 and height 9.

Solution



Total SA = two circular bases + the lateral (curved) surface.

Two circles plus the curved lateral surface: $SA = 2\pi r^2 + 2\pi rh = 2\pi r(r + h)$.

$SA = 2\pi(4)(4 + 9) = 2\pi(4)(13)$

$SA = 104\pi$.

Answer: 104π square units (exact form — leave π in the answer).

Example 4: Square pyramid — slant height given

A square pyramid has base edge 10 and slant height 13. Find its total surface area.

Solution

Base area: $B = s^2 = 10^2 = 100.$

Perimeter: $P = 4s = 40.$

Lateral area (4 congruent triangles) $= \frac{1}{2} P \ell = \frac{1}{2}(40)(13) = 260.$

$SA = B + \frac{1}{2} P \ell = 100 + 260 = 360.$

Answer: 360 square units.

Example 5: Square pyramid — find the slant height first

A square pyramid has base edge 10 and (vertical) height 12. Find its total surface area.

Solution

The slant height ℓ is *not* the height h — it runs from the apex down the middle of a triangular face to the midpoint of a base edge. It is the hypotenuse of a right triangle with legs h and half the base edge:

$\ell = \sqrt{h^2 + \left(\frac{s}{2}\right)^2} = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13.$

Now use $\ell = 13$ exactly as in Example 4: $B = 100, P = 40.$

$SA = 100 + \frac{1}{2}(40)(13) = 100 + 260 = 360.$

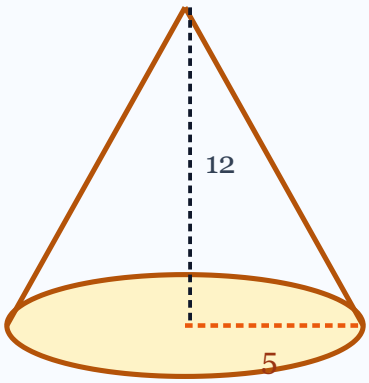
Answer: 360 square units. (Same numbers as Example 4 — this shows how $h = 12$ and $\ell = 13$ are connected.)

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Example 6: Cone — find the slant height first

A cone has radius 5 and height 12. Find its total surface area.

Solution



The slant height ℓ is the hypotenuse of the right triangle formed by r and h.

Find ℓ using the Pythagorean Theorem: $\ell = \sqrt{r^2 + h^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

$SA = \pi r^2 + \pi r \ell = \pi(25) + \pi(5)(13) = 25\pi + 65\pi$

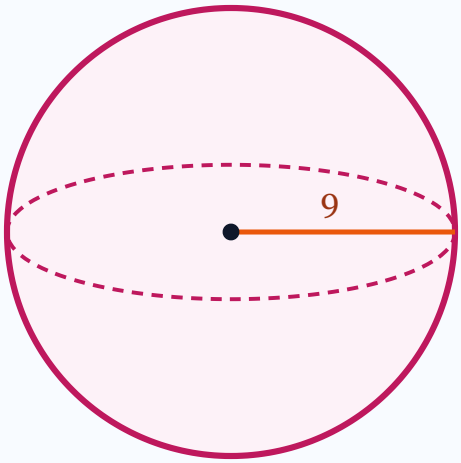
$SA = 90\pi$.

Answer: 90π square units.

Example 7: Sphere

A sphere has diameter 18. Find its surface area.

Solution



Remember: the sphere formula uses the radius, not the diameter.

Radius: $r = \dfrac{18}{2} = 9$.

$SA = 4\pi r^2 = 4\pi(81) = 324\pi$.

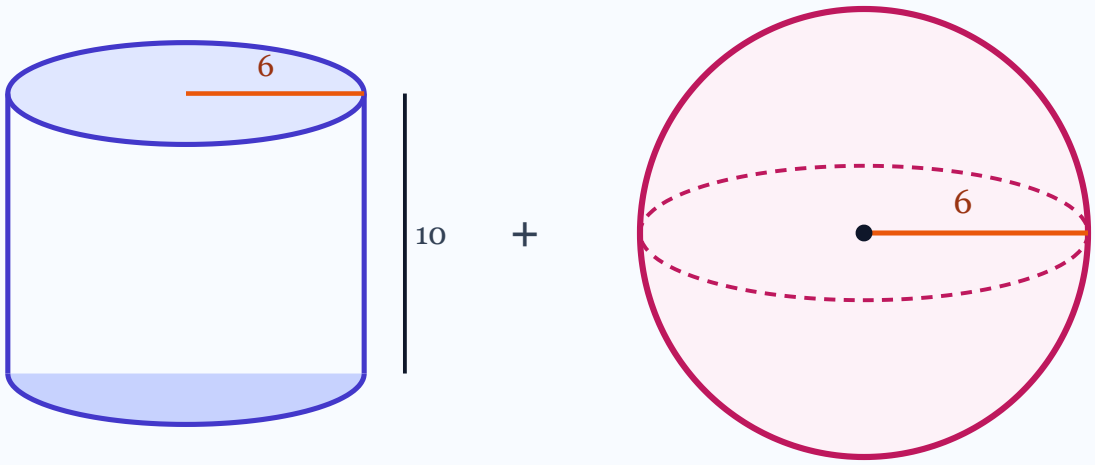
Answer: 324π square units.

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Example 8: Composite solid — silo (cylinder + hemisphere)

A grain silo is a cylinder of radius 6 and height 10 topped by a hemispherical dome of the same radius. Find the total *outer* surface area (the silo sits on the ground, so ignore the bottom, and ignore the seam where the dome meets the cylinder).

Solution



Composite outer surface = cylinder's lateral area + hemisphere's curved area. No bottom circle, no flat seam circle (it's internal).

Step 1 — cylinder's lateral (curved) surface only (no top, no bottom, since the dome covers the top and the silo sits on the ground):

$SA_{\text{cyl, lateral}} = 2\pi rh = 2\pi(6)(10) = 120\pi.$

Step 2 — hemisphere's curved surface only (half of a full sphere's SA, no flat circle since it's fused to the cylinder):

$SA_{\text{hemi}} = \frac{1}{2}(4\pi r^2) = 2\pi r^2 = 2\pi(36) = 72\pi.$

Step 3 — add the pieces: $SA_{\text{total}} = 120\pi + 72\pi = 192\pi.$

Answer: 192π square units.

Example 9: Lateral SA vs. total SA

A cylinder has radius 5 and height 8. (a) Find its lateral surface area. (b) Find its total surface area.

Solution

(a) Lateral SA is just the curved side — no circles: $SA_{\text{lat}} = 2\pi rh = 2\pi(5)(8) = 80\pi$.

(b) Total SA adds both circular bases: $SA_{\text{total}} = 2\pi rh + 2\pi r^2 = 80\pi + 2\pi(25) = 80\pi + 50\pi = 130\pi$.

Answers: lateral $= 80\pi$; total $= 130\pi$. *The difference, 50π, is exactly the area of the two circles.*

Example 10: Reverse problem — solve for a missing dimension

A rectangular prism has length 10, width 6, and total surface area 280. Find its height.

Solution

Set up the SA formula with h unknown:

$SA = 2(lw + lh + wh) = 2(10(6) + 10h + 6h) = 2(60 + 16h) = 120 + 32h$.

Set equal to the given SA: $120 + 32h = 280$

$32h = 160$

$h = 5$.

Check: $2(60 + 50 + 30) = 2(140) = 280$. ✓

Answer: $h = 5$.

Example 11: Open-top tank (no lid)

A cylindrical water tank with no lid has radius 3 and height 10. Find the surface area of material needed to build it (walls + bottom only).

Solution

Only one circular base is needed (the bottom); the top is open:

$SA = \pi r^2 + 2\pi rh = \pi(9) + 2\pi(3)(10) = 9\pi + 60\pi$

$SA = 69\pi$.

Answer: 69π square units. *Compare to a closed tank, which would need an extra 9π for the lid.*

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Rectangular prism

Find the SA of a rectangular prism with length 6, width 4, height 5.

Answer: **Verification:**

Problem 2: Cube

Find the SA of a cube with edge 9.

Answer: **Verification:**

Problem 3: Cylinder

Find the total SA of a cylinder with radius 5, height 11.

Answer: **Verification:**

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Problem 4: Pyramid (slant given)

Square pyramid, base edge 8, slant height 9. Find total SA.

Answer: **Verification:**

Problem 5: Cone (slant given)

Cone with radius 9, slant height 15. Find total SA.

Answer: **Verification:**

Problem 6: Sphere from radius

Find the SA of a sphere with radius 7.

Answer: **Verification:**

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Problem 7: Sphere from diameter

Find the SA of a sphere with diameter 20.

Answer: **Verification:**

Problem 8: Cone — find ℓ first

Cone with radius 8, height 6. Find the slant height, then the total SA.

Answer: **Verification:**

Problem 9: Pyramid — find ℓ first

Square pyramid, base edge 14, height 24. Find the slant height, then the total SA.

Answer: **Verification:**

Problem 10: Composite silo

A silo is a cylinder (radius 4, height 9) topped with a hemisphere of the same radius. Find the total outer SA (no bottom, no seam).

Answer: **Verification:**

Problem 11: Reverse problem

A rectangular prism has length 8, width 5, and SA 262. Find its height.

Answer: **Verification:**

Problem 12: Open-top cylinder

An open-top cylindrical tank has radius 6, height 12. Find the SA of material needed (no lid).

Answer: **Verification:**

Problem 13: Concept

Explain why a square pyramid's total SA formula adds only *one* base area, while a prism's formula effectively adds two.

Answer: **Verification:**

Problem 14: Word problem

A silo (cylinder radius 5, height 14, topped by a hemisphere of the same radius) needs to be painted on the outside, excluding the bottom (in the ground). Find the area to be painted.

Answer: **Verification:**

Problem 15: Challenge

A cone and a cylinder share the same base radius r. The cylinder has height h and the cone has slant height ℓ. If their total surface areas are equal, show that $\ell = r + 2h$.

Answer: **Verification:**

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✓ Answer Key

1. 148
2. 486
3. 160π
4. 208
5. 216π
6. 196π
7. 400π
8. $\ell = 10$; $SA = 144\pi$
9. $\ell = 25$; $SA = 896$
10. 104π
11. $h = 7$
12. 180π
13. A pyramid has a single base; a prism has two congruent parallel bases
14. 190π
15. $2\pi r(r + h) = \pi r(r + \ell) \Rightarrow 2(r + h) = r + \ell \Rightarrow \ell = r + 2h$

Great work — keep practising!